Weekly Progress: Optimizing Multilingual Chatbot Output

On Development of an on-Premise AI Product Recommendation Model Using Open-Source LLM

Atique Md Monir Ahammod Bin (248810)
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Department of Artificial Intelligence Convergence

SOFTWARE & SYSTEMS
LABORATORY FOR SECURITY

전남대학교
CHONNAM NATIONAL UNIVERSITY

Project Description

Project Title

Development of an on-Premise AI Product Recommendation Model Using Open-Source LLM

Company: Narosu Co. Ltd.

Project Description

- The e-commerce market is becoming increasingly competitive, requiring advanced recommendation systems.
- Large marketplaces use sophisticated AI models, but independent online store operators lack such solutions.
- This project aims to develop an **AI-powered product recommendation system** tailored for self-managed online stores.
- By analyzing customer preferences, the system will suggest products that align with their interests.

Project Objective

- Develop an AI-based search and recommendation system that analyzes user input sentences and recommends the **TOP 5** most relevant products based on semantic similarity.
- Utilize Natural Language Processing (NLP) and vector search technology to provide a highly refined search experience.
- Deploy in an on-premise environment to ensure data privacy and full control over the recommendation process.

Company Requirements

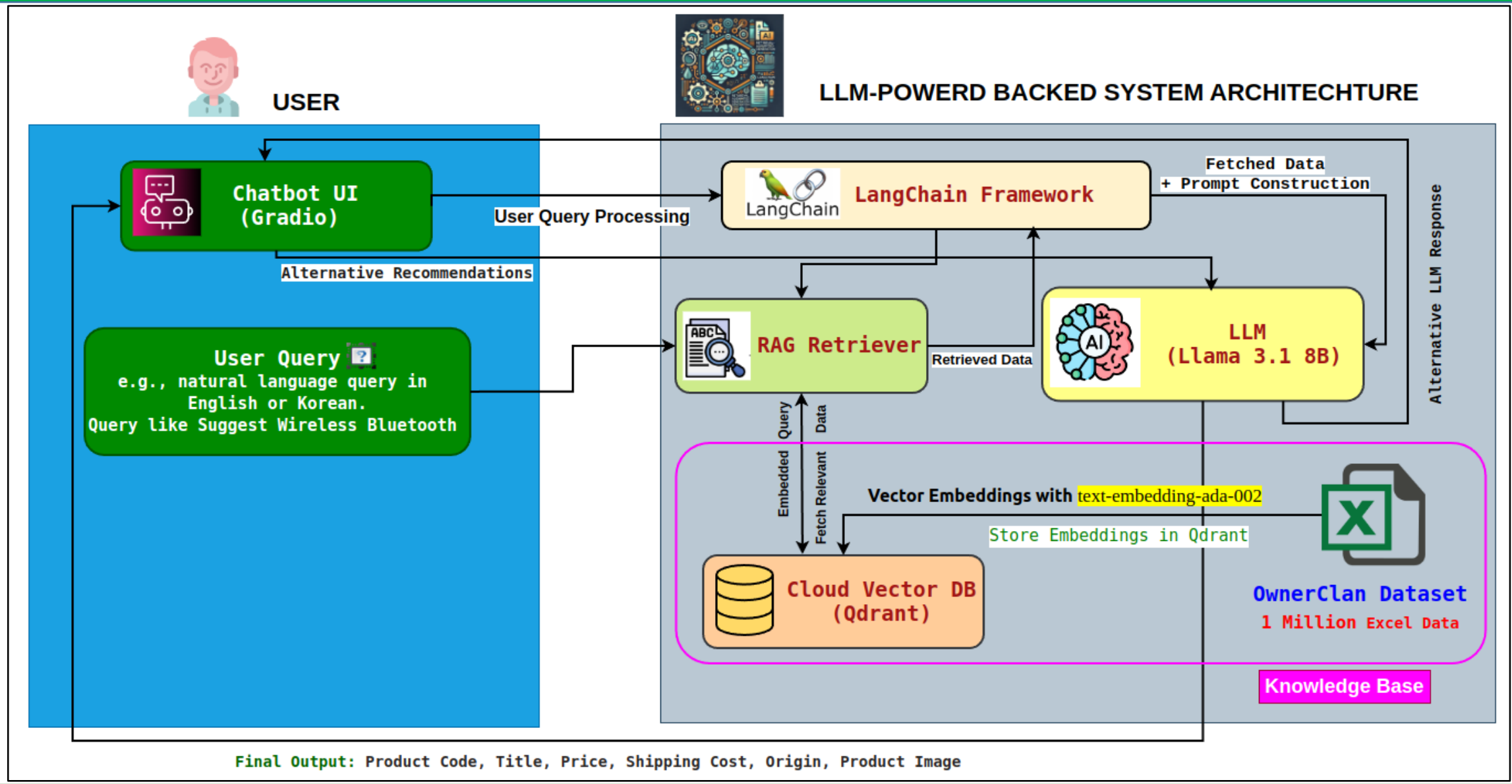
Key Technology Stack

Component	Technology Used	Description
LLM (Large Language Model)	Llama 3.1 8B	Understands and generates natural language based on context
LangChain (LLM Framework)	LangChain	Enhances AI responses by connecting LLM to external data sources
Embedding Model	OpenAI text-embedding-ada-002	Generates text embeddings based on semantic similarity
Vector DB	Flexible (Any compatible DB)	Stores high-dimensional vector data
Vector Search Library	FAISS	Optimized large-scale vector data search
RAG (Retrieval-Augmented Generation)	RAG-based search + LLM	Retrieves relevant documents in real time to generate accurate answers
API Architecture	REST API (FastAPI, Flask, or Django)	Builds an API for client-server data communication
GUI (Graphical User Interface)	Chatbot	Applies a chatbot template for a user-friendly and intuitive interface

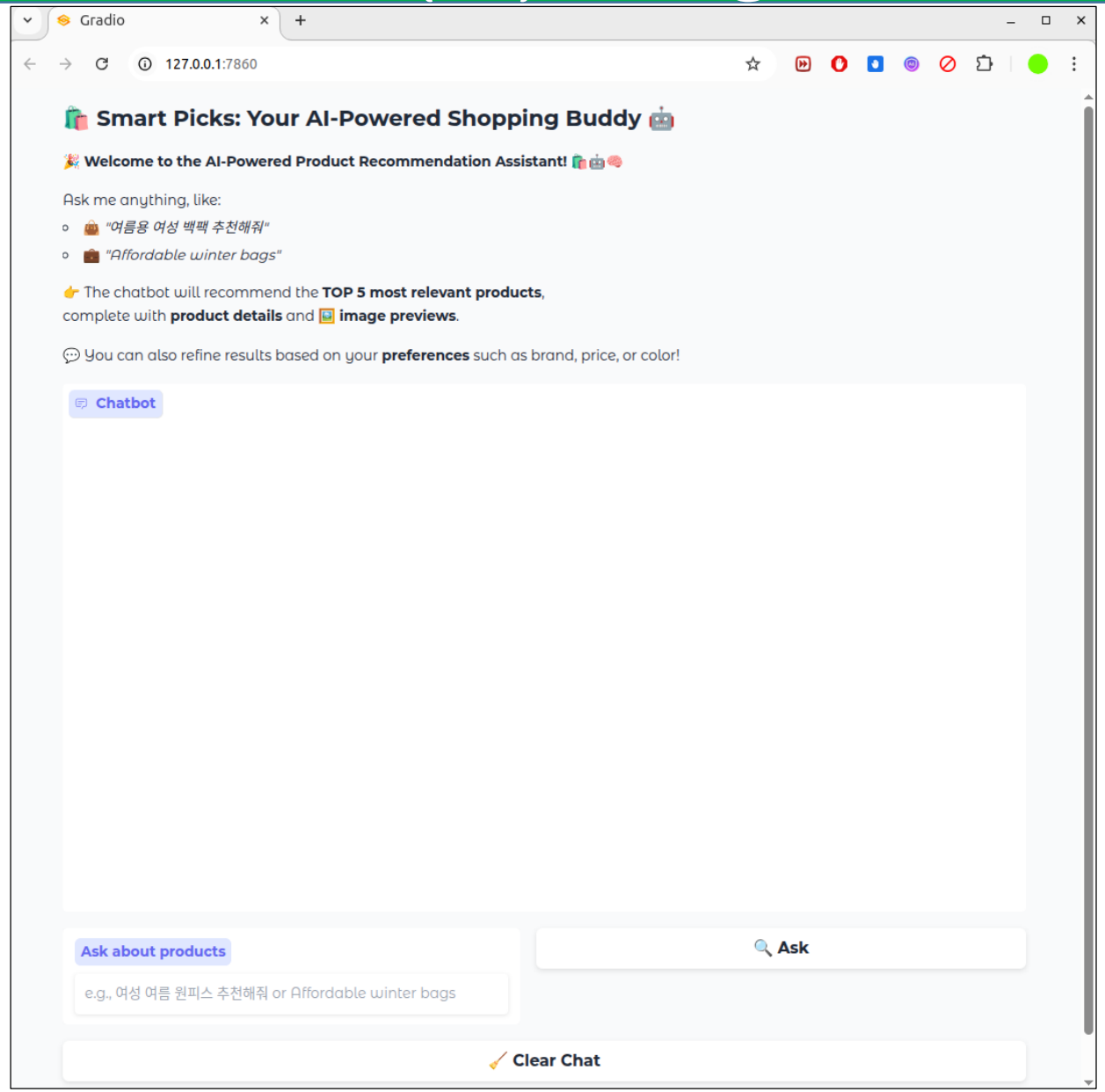
Selected Model and Vector Store:

- **Embedding Model**: `text-embedding-ada-002`
- **Vector DB/Search:** Qdrant
- **LLM Reasoning:** Llama 3.1 8B instant
- **LangChain Framework for LLM**
- **UI Framework: Gradio**
- **API Architecture: Gradio + FastAPI**

System Configuration (AI Model) Diagram



ChatBot User Interface (UI) Using Gradio



Project Schedule (Gantt Chart)

Phase	2025													
	March				April				May				June	
	1	2	3	4	1	2	3	4	1	2	3	4	1	2
Project Understanding and Data Collection														
Data Analysis and Pre-processing														
Model Selection and Training														
System (RAG) Development														
Chatbot Development														
Testing and Deployment														

Previous Communication with Company

Asked some queries to the company:

전남대학교 AI융합프로젝트 나유경입니다.

요청하신 **기술스택** 정리하여 보내드립니다. (sheet 별로 정리되어 있습니다)
첨부 : 김성준+나유경+Monir_Technology_Stack.xlsx

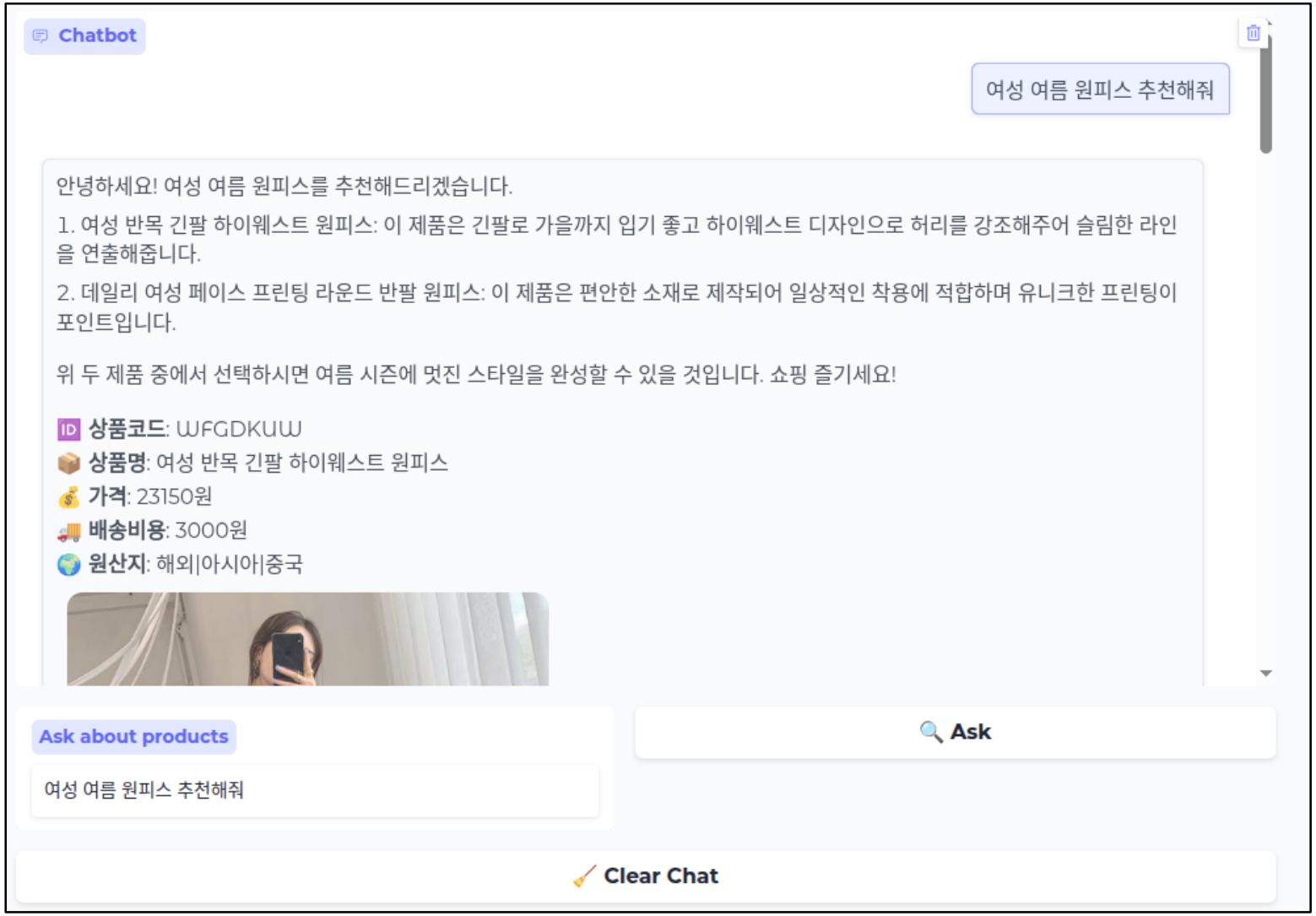
또한, **문의사항** 몇 가지 드립니다.

1. Streamlit과 같은 데모용 프레임워크를 이용해 프론트와 백엔드를 진행해도 될까요?
2. 이전의 요구사항에서 Naver의 상품이 추천될 수 있도록 챗봇을 만들라고 하였는데, 혹시 꼭 진행해야하는 부분일까요? 마감기한이 다소 타이트합니다.
3. 짧은 시간 내에 어렵지만, 동시 접속하는 여러 사용자를 고려하여 챗봇을 구현해야할까요?

항상 수고 많으십니다. 감사합니다.
나유경 드림.

Testing the Chatbot with Korean Query

1. User Query in Korean: 여성 여름 원피스 추천해줘

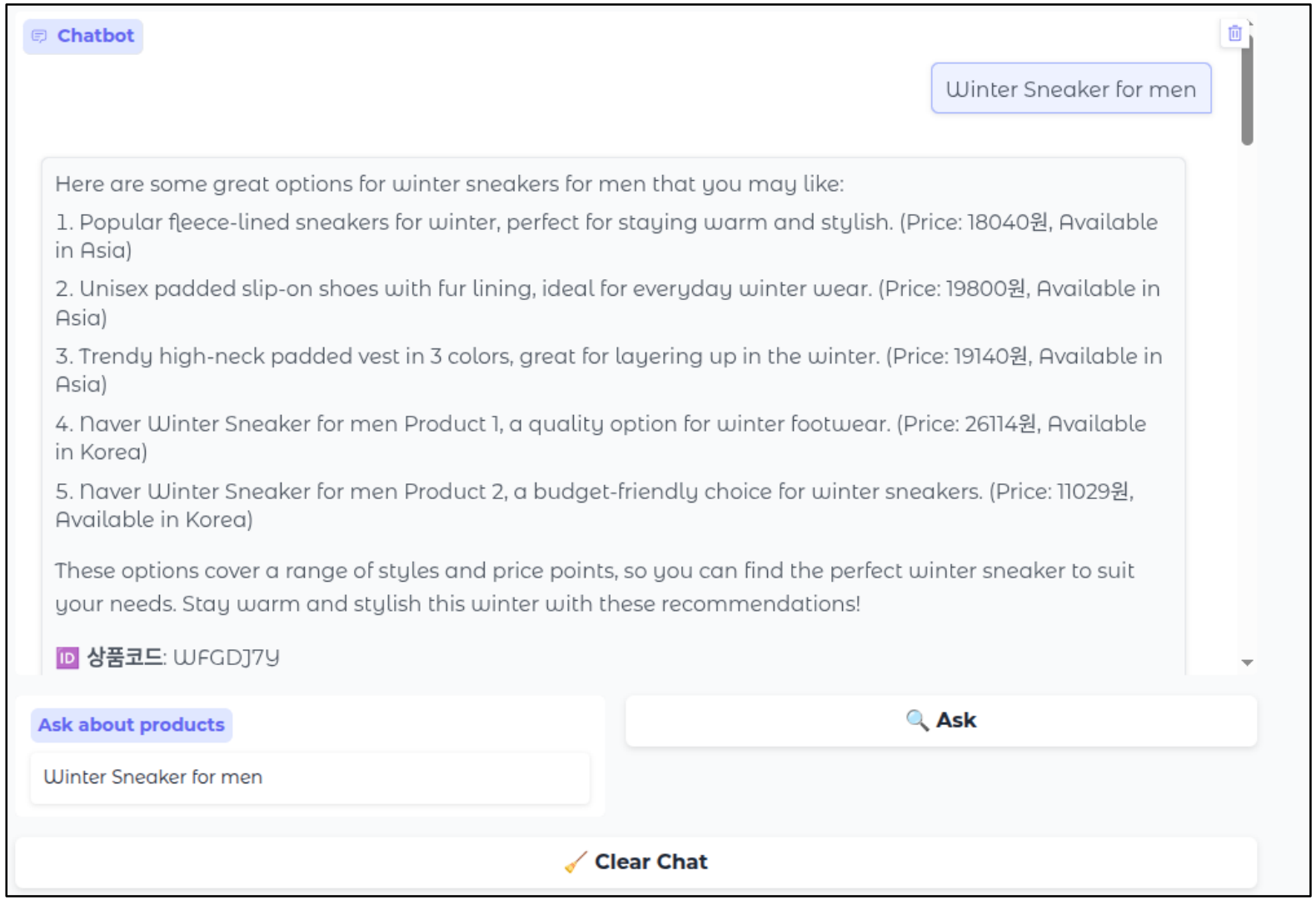


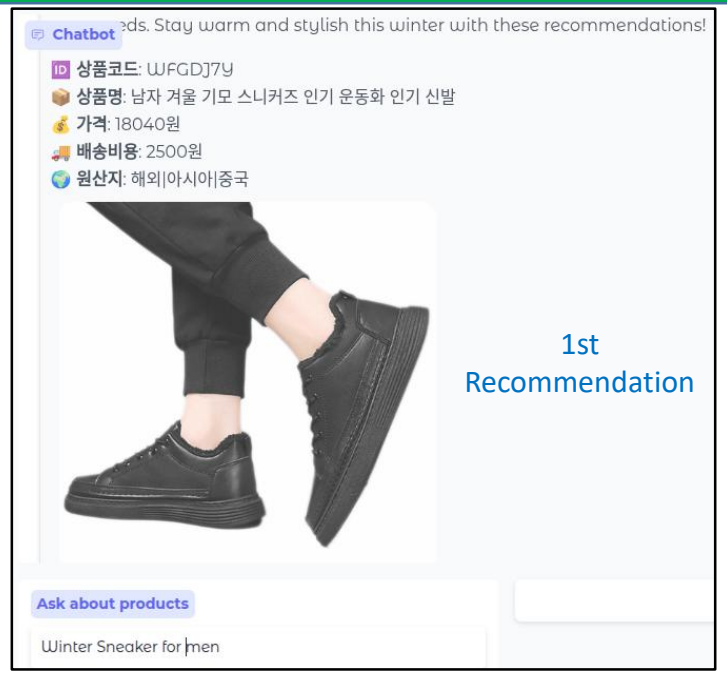
Recommendation in Korean is Okay



Testing the Chatbot with English Query

1. User Query in English: Winter Sneaker for men





1st Recommendation



2nd Recommendation

Need to translate the product details in English

Code for Multilingual Chatbot Output

1. Import langdetect and deep_translator library.

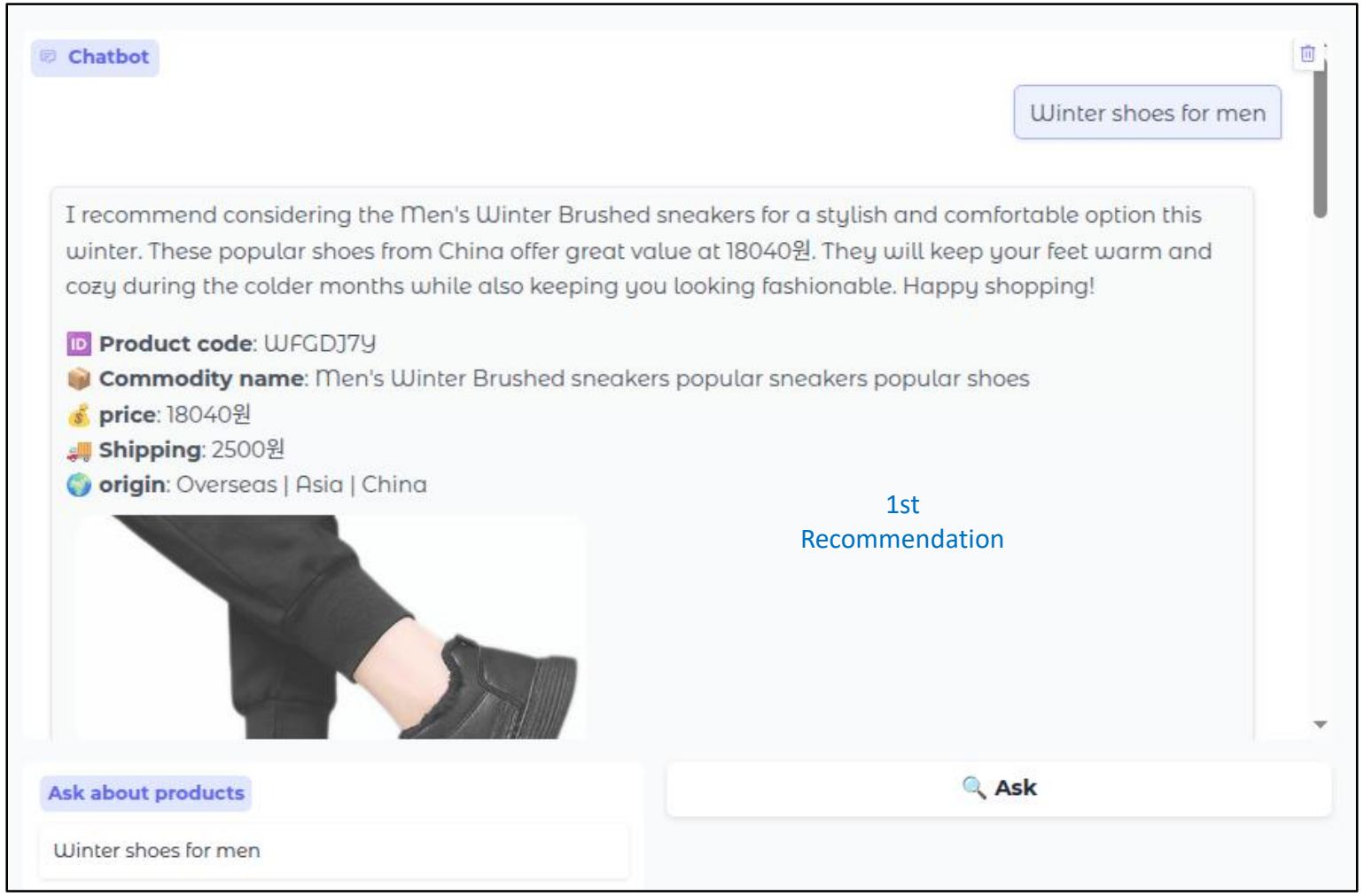
```
from langdetect import detect
from deep_translator import GoogleTranslator
```

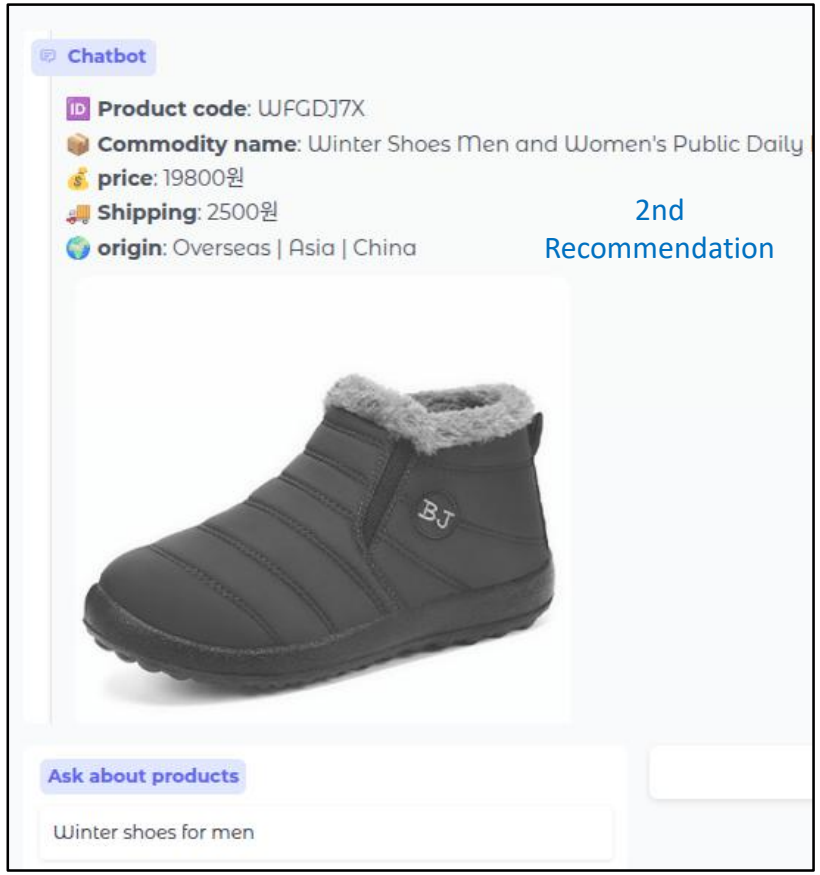
2. A helper function to translate product details into user query language.

```
# -------------------- Helper function to translate product info for LLM--------------------
def translate_product_info(product_infos, target_lang):
    if target_lang == "ko":
        return product_infos
    translated_products = []
    for p in product_infos:
        translated = {}
        for k, v in p.items():
            if k in ["상품코드", "이미지대URL"]:
                translated[k] = v
            else:
                try:
                    translated[k] = GoogleTranslator(source='ko', target=target_lang).translate(v)
                except Exception:
                    translated[k] = v  # fallback in case translation fails
        translated_products.append(translated)
    return translated_products
```

Multilingual Chatbot with English Query

1. User Query in English: Winter shoes for men

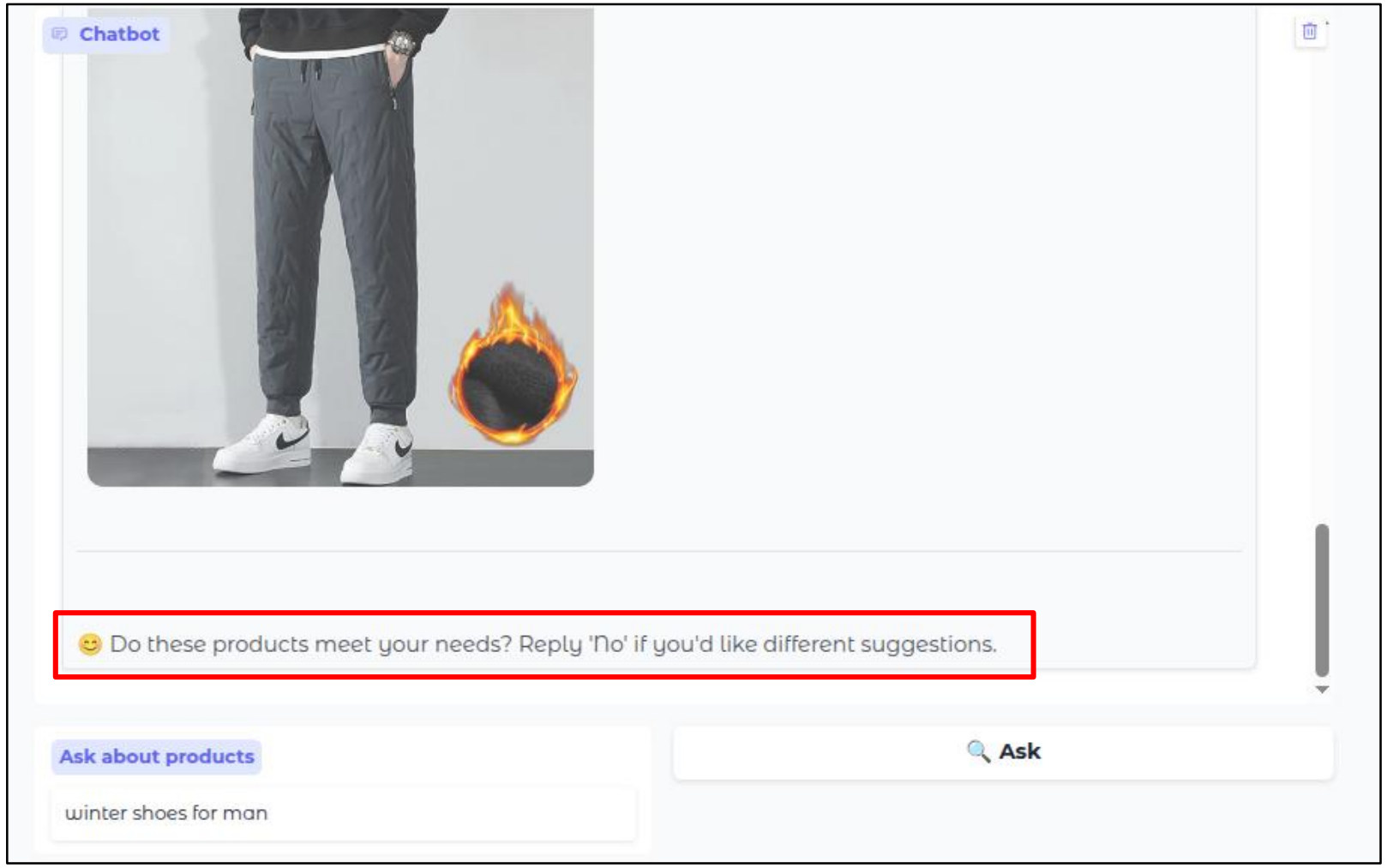




Now the recommendation in user query language

Alternative Recommendations

1. Asking for an alternative suggestion



Next Tasks To Do

Next Step

- Support alternative suggestions
- Upgrade the Chatbot UI for showing a user profile and login feature.

Thank You for Listening.